Customer Segmentation Analysis Report

RFM Framework & Unsupervised K-Means Clustering Strategy

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Methodology:	RFM (Recency, Frequency, Monetary), K-Means Clustering, StandardScaler	**Date:**	June 25, 2026

1. Executive Summary & Objective

The primary objective of this analysis is to identify distinct customer behavioral segments within the retail transactional ecosystem using data-driven unsupervised learning. By evaluating when customers last purchased (**Recency**), how often they buy (**Frequency**), and the total economic value they generate (**Monetary**), we transition generic customer data into granular actionable marketing buckets.

Using a cohort of **4,338 unique customers** derived from cleaned online retail transactions, a 4-cluster K-Means algorithm was deployed to reveal non-obvious segments ranging from high-value champions to dormant accounts at high risk of churn.

2. Data Preparation & Feature Engineering

Raw data inputs from `OnlineRetail.csv` underwent a rigorous engineering workflow to ensure model stability:

- **Data Auditing & Cleaning:** Rows with missing `CustomerID` and description records were pruned. Negative entries in `Quantity` and `UnitPrice` representing cancellations or corrections were removed to focus explicitly on real revenue behavior.
- **Feature Engineering:** Total transaction spend was structured via `TotalAmount = Quantity × UnitPrice`.
- **RFM Metrization:** A reference snapshot date was set exactly one day past the maximum invoice timestamp (`2011-12-10`). Individual records were compiled per unique customer identifier to fetch standard metrics.
- **Z-Score Normalization:** Because K-Means relies heavily on Euclidean distance calculations, metrics were scaled using `StandardScaler` to mitigate skewness and scale disparities.

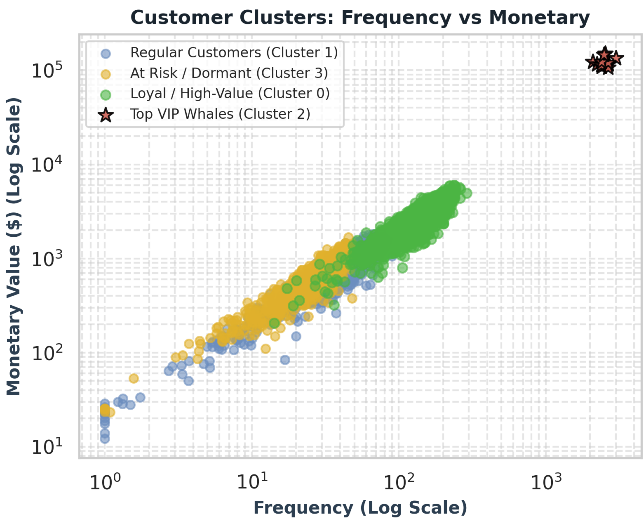
3. Customer Cluster Performance Profiles

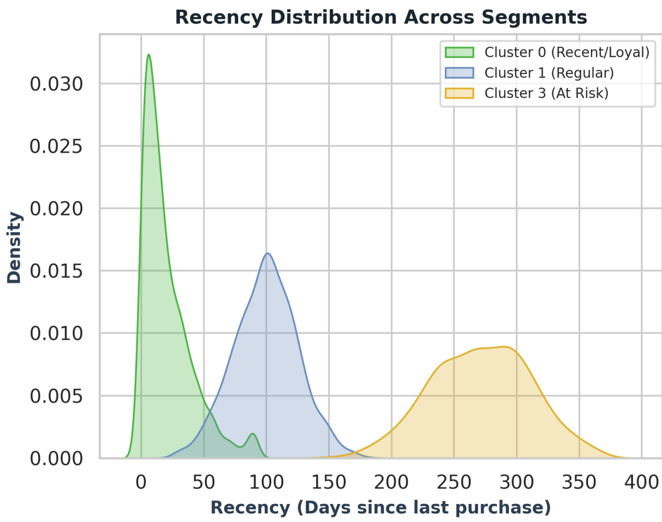
The table below provides the mathematical centroid profiles for the four identified segments:

Cluster Index	Segment Assignment	Customer Volume (%)	Avg. Recency (Days)	Avg. Frequency (Orders)	Avg. Monetary Value ($)
Cluster 0	**Loyal & High-Value Customers**	2,171 (50.0%)	20.99	135.29	$2,645.82
Cluster 1	**Mid-Tier Regulars**	1,326 (30.6%)	98.22	37.72	$773.88
Cluster 2	**Top VIP / "Whales"**	13 (0.3%)	4.69	2,565.31	$126,118.31
Cluster 3	**Dormant / At-Risk Churn**	828 (19.1%)	272.41	25.14	$605.84

4. Visual Segmentation Analytics

The following visualizations illustrate how the K-Means algorithm partitioned behavior across scales, utilizing logarithmic coordinate mapping to safely isolate extreme high-value spending patterns without masking mid-market metrics:





5. Operational Interpretations & Marketing Actions

💎 Cluster 2: The Top VIP Whales (0.3% of Base)

Despite comprising only 13 accounts, this group holds massive economic influence, averaging over **$126k in value** with a recency metric of under 5 days.
Strategy: Hyper-personalized exclusive loyalty pipelines, personal account managers, and immediate sneak-peeks at new merchandise updates.

📈 Cluster 0: Loyal & High-Value Customers (50.0% of Base)

The bedrock core of the business revenue model. Highly active (avg. 135 orders) and consistently purchasing within a 3-week window.
Strategy: Implement cross-selling models, milestone-based value rewards, and referral programs to turn them into brand advocates.

🕒 Cluster 1: Mid-Tier Regulars (30.6% of Base)

Steady occasional buyers whose frequency metrics lag behind their loyal peers.
Strategy: Deploy time-sensitive trigger promotions, customized newsletters, and low-threshold multi-buy discount triggers to raise purchase velocity.

⚠️ Cluster 3: Dormant / Churn Threat (19.1% of Base)

Accounts that have largely flatlined with an average inactivity span exceeding 9 months (272 days).
Strategy: Aggressive win-back campaigns ("We Miss You"), high-incentive standard value vouchers, and direct consumer satisfaction surveys.

Conclusion & Data Sign-off

By replacing baseline historical aggregates with dynamic K-Means clusters, marketing teams can optimize budget allocation towards groups yielding the highest relative Customer Lifetime Value (CLV). This report concludes the technical data assignment for Task 11.